

VOLUME I — TECHNICAL

Frasier Digital, LLC

RFQ 75D301-26-Q-00146 · Modernized VAERS Reporting Application

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Prototype for evaluation: <https://vaers-demo.frasierdigital.com>

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Tab 1 — Executive Summary

Frasier Digital, LLC — a Texas small disadvantaged business — proposes to design, build, and deploy the modernized VAERS reporting application described in RFQ 75D301-26-Q-00146, within CDC's FedRAMP-authorized Azure environment, open source per M-16-21, Section 508-conformant, delivered by a named principal-led team with deep healthcare-IT experience (Optum/UnitedHealthcare, healthtech platform engineering, 60k-user healthcare systems).

We did not wait for award to demonstrate understanding. A working prototype of the modernized reporting experience — branching public/provider form on the actual VAERS 2.0 data elements, provider vaccine-error path with adverse-event field suppression, plain-language toggle, guarded intelligent completion assistance, low-code CDC-editable administration, and VAERS-compatible structured output — is live today for evaluation at **<https://vaers-demo.frasierdigital.com>** (Tab 2-2), tested to 0 WCAG 2.1 AA violations.

Our quote includes every required gate: the Data Management Plan, the completed AI Use Compliance and Risk Management Plan (Attachment 1 template), the Accessibility Conformance Report, itemized pricing, and signed representations. Firm-fixed-price CLIN 0001 with travel CLINs at the Government's stated NTE amounts; no other direct costs — the stack is open source and the cloud is Government-furnished.

The team named in this proposal is the team that will do the work.

Tab 2-1 — Technical Plan

1. Understanding of the Requirement

VAERS is the nation's front door for vaccine safety signals, and that front door is hard to walk through. A parent reporting a child's reaction and a pharmacist reporting a vaccine error today face the same experience: a long, expert-oriented form served by a legacy ColdFusion/Java stack, a writable PDF alternative, and navigation that predates modern mobile use. The predictable results are the ones this RFQ targets: abandoned submissions, incomplete critical fields, and provider reports forced through adverse-event questions that do not apply. Every abandoned report is a data point the Immunization Safety Office never sees.

CDC is not buying a website refresh. It is buying **data quality and completeness at the point of capture**, delivered as a modern reporting experience: branching logic that asks each submitter only what applies (Task 1.6), plain language for the public path (1.6.1), AE-field suppression for provider vaccine-error reports (1.6.2), intelligent completion assistance (Section 2), integrated medical-record upload with a disciplined Phase 1 scope (Task 2), and VAERS-compatible structured outputs that feed existing downstream systems without disrupting them (Task 1.9) — all inside CDC's FedRAMP-authorized Azure environment, under an ATD CDC can defend, conforming to Section 508, and editable by CDC program personnel without a developer in the loop (Task 1.8).

Three constraints shape our approach. **First, the new application is a guest in a working ecosystem:** existing VAERS support contracts, downstream data workflows, CXone and eFax channels, and the post-submission upload tool all continue operating; our integration surface is VAERS-compatible structured data and coordination through CDC, not replacement of back-end systems. **Second, the nine-month period of performance with fixed deliverable clocks (wireframes DOA+21, alpha DOA+45, beta DOA+90, production DOA+150) leaves no room for a discovery phase that produces only documents** — which is why we arrive with a working, tested prototype (Tab 2-2) rather than a promise of one. **Third, trust requirements are asymmetric:** a public submitter must never be given medical advice or a causality opinion by an assistance feature, and PHI/PII discipline must be engineered in from the first wireframe, not audited in at the end.

We have already reduced this understanding to working software. The prototype at **vaers-demo.frasierdigital.com** (Tab 2-2) implements the branching form on the actual VAERS 2.0 data elements, the provider vaccine-error path with AE suppression, the plain-language/clinical toggle, completion assistance with guardrails, the low-code admin surface, and VAERS-compatible structured output — evaluated at 0 WCAG 2.1 AA violations across every page state. Every technical claim in this tab can be inspected there today.

2. Technical Approach and Architecture

Architecture in one paragraph. A responsive single-page web application (React + TypeScript, USWDS-based design system) served from CDC's Azure environment, backed by a lightweight API tier (Node.js) and Azure SQL within the CDC tenancy. The submission form is **schema-driven:** form structure, branching rules, field text, tooltips, and help content live in versioned configuration, not code — this single design decision is what makes the low-code interface (Task

1.8) real rather than aspirational, because CDC program personnel edit the same configuration the form engine reads. Intelligent completion assistance runs against **CDC's existing enterprise Azure OpenAI service (EDAV)** behind a deterministic safety gate — no new AI services, no data leaving the boundary (our completed AI Use Compliance and Risk Management Plan, Attachment 1, governs this in detail). All custom code is delivered open source under M-16-21 to a CDC repository on a bi-weekly cadence. **[FLAG: final stack wording pending Bish Q&A on EDAV availability; prototype covers both smart-validation-only and assistant answers.]**

Why this is sound. Every load-bearing element is boring on purpose: React and Node are already present in CDC's VAERS environment; Azure SQL MI is the incumbent data platform; USWDS is the federal design standard; the AI layer is CDC's own service. The innovation is concentrated where it pays — branching schema engine, assistance guardrails, low-code editing — and the risk is concentrated where we can retire it early, which the schedule below does.

Scalability and Phase 2 posture. The schema-driven engine scales sideways (new submitter types, new form versions, Spanish-language variant) without re-architecture. The upload subsystem is built with Phase 1 restrictions enforced in configuration, so Phase 2 expansion to images (2.4.2) is a policy change plus review, not a rebuild. Stateless application tier scales horizontally under Azure load balancing; surge events (a safety communication driving reporting spikes) are a capacity configuration, not an engineering project.

3. Approach by PWS Task

Task 1 — Design and Develop the Web Application

1.1–1.2 Requirements confirmation and UX artifacts. Weeks 1–3 are structured stakeholder sessions with the VAERS program office to confirm data elements, integration constraints, and success metrics — anchored on our prototype as the strawman. Reviewing working software instead of static wireframes gets CDC to real decisions ("suppress these three fields on this branch," "this tooltip is wrong") in the first session. UX deliverables (user flows, wireframes, clickable prototype, landing/navigation redesign) are due DOA+21; we hit that clock by iterating the existing prototype under CDC direction, and we capture the abandonment-rate baseline methodology with CDC at kickoff (feeding 1.10/PRS#4).

1.3 Responsive application. Mobile-first build against a Government-agreed device/browser matrix (PRS#2), with field-level validation, inline error recovery, and save-and-resume so an interrupted submitter does not become an abandoned report. Automated cross-viewport regression tests run in CI on every commit.

1.4 Landing page and navigation redesign. Three clear front doors — report an adverse event, check on a report, find data — with reporting tools, FAQs, and data downloads reachable in ≤ 2 clicks (PRS#3). Redesign ships in the DOA+21 UX package.

1.5 / 1.7 Satisfaction surveys. Two instruments: site-navigation survey and post-submission pop-up (focus-trapped, 508-conformant, dismissible without penalty). Instruments delivered with Beta; response data reported per the approved schedule (PRS#7) and summarized in monthly status reports.

1.6 Branching-logic submission form. The core of the job. One schema, two rendered experiences: a plain-language public path written to a general reading level (1.6.1) and a clinical

provider path — with branching that presents and suppresses fields per submitter type and report characteristics, including the mandated provider vaccine-error-no-AE path (1.6.2, PRS#1).

Branching rules are table-driven and unit-tested against a scenario matrix we expect the Government to extend during acceptance; the prototype already passes the vaccine-error-no-AE scenario, visibly announcing "N questions removed" so submitters trust the shortening form.

1.8 Low-code/configurable interface. Authorized CDC personnel edit field text, tooltips, help/FAQ content, choice options, and field visibility through an administrative interface with live preview, role-based access, versioning, and one-click rollback — no developer, no deployment (PRS#8). We will jointly script the Government's update scenarios early so acceptance testing runs against rehearsed reality. The change and configuration process is documented as part of Task 1.8 and taught to program personnel in a hands-on session before Beta.

1.9 VAERS-compatible data capture and transmission. Structured outputs mapped field-for-field to the VAERS data dictionary (the prototype already emits VAERSDATA/VAERSVAX-shaped records including derived flags and computed fields), with transmission mechanics (format, channel, cadence) confirmed with CDC and the existing system contractors during 1.1 and proven in Government acceptance testing (PRS#6). De-identification boundaries mirror the published public-data split.

1.10 Performance. ≤ 3 s page loads, ≤ 10 min median submission, $\geq 30\%$ abandonment reduction versus the kickoff baseline — engineered (static-first delivery, code splitting, CDN caching per CDC architecture) and then **measured and reported** with an approved methodology (PRS#4): real-user timing metrics, funnel analytics on abandonment, monthly reporting with deviations explained. The current prototype loads in ~ 1.2 s on commodity hosting.

1.11 Iterative builds. Working builds delivered for Government review from Alpha (DOA+45) onward on a bi-weekly cadence aligned to technical reviews; defect triage with severity SLAs runs continuously through Beta (DOA+90) and Final (DOA+150).

1.12 User documentation. Public, provider, and administrator guides written against the final UX (PRS#16), plain-language reviewed, 508-conformant, delivered with Final.

Task 2 — Medical Record Upload and Document Management

Integrated upload lands in the form flow (2.1) while the existing post-submission tool continues untouched — its users lose nothing. A free-text box (2.2) captures narrative detail the structured fields miss. The **supplemental-document suggestion tool (2.3)** scans provider submissions against a transparent, configurable rule set (report characteristics → suggested document types) and reaches $\geq 90\%$ suggestion coverage across Government-defined scenarios (PRS#10); the rules live in the same low-code configuration as the form, so CDC can tune suggestions as reporting patterns evolve. **Phase 1 restrictions (2.4.1) are enforced server-side** — accepted document categories are configuration, giving CDC a designed, documented Phase 2 expansion path (2.4.2, PRS#11) without a rebuild. All upload handling is PHI-grade: encryption in transit and at rest, type/size validation, malware scanning per CDC standards, audit logging, least-privilege access (2.5, PRS#9). Defect remediation and upload performance optimization continue through Beta and Final (2.6).

Task 3 — Security, Privacy, Compliance, and ATD

Compliance runs as an engineering workstream from day one, not a documentation phase at the end. **Privacy by architecture (3.1):** PII/PHI minimization, no identifiers required for assistance features, de-identification split mirrored from VAERS public-data practice, HIPAA/Privacy Act protections documented in the PTA/PIA path. **CDC environment only (3.2):** all development, testing, and deployment inside the CDC-managed FedRAMP Azure boundary; we coordinate environment stand-up with CDC/OCIO and existing VAERS hosting teams in month one. **ATD package (3.3, PRS#12):** security and privacy documentation drafted with Beta and submitted in advance of deployment, with a named owner (PM) for Government review comments and remediation — our schedule holds a dedicated remediation window so ATD review does not sit on the critical path unbuffered. **Safeguards (3.4, PRS#13):** access control, FIPS-validated encryption, audit logging, secure configuration baselines, and vulnerability scanning in CI during development plus pre-deployment scans with zero known critical/high findings at deployment. **Section 508 (3.5, PRS#14):** automated accessibility testing on every commit plus manual assistive-technology testing each release; ACR/VPAT delivered with Beta and updated at Final — the prototype's 0-violation axe results across all page states show this is our default engineering posture, not a remediation activity. Compliance documentation package (3.6, PRS#15), functional/security/accessibility test-and-remediate cycles before Beta and Final (3.7), and administrator/technical guides including Task 1.8 configuration guidance (3.8, PRS#16) complete the task.

Task 4 — Transition-Out

Transition readiness is a standing condition, not an end-of-contract scramble: because code goes to CDC's repository bi-weekly under M-16-21 with documentation maintained in the same repo, the Government holds current source, configuration, and process assets at all times. We assess and report transition readiness monthly to the COR (4.1, PRS#18), deliver the detailed Transition-out Plan at least two months before contract end (4.4), lead TEMs and shadowing with any successor (4.5, 4.10), and deliver final documentation, asset inventory reconciliation, account/access handover, and closeout (4.2–4.3, 4.6–4.11) with no degradation of support through the final day.

4. Schedule and Milestones

Clock (from DOA)	Milestone / Deliverable	Our internal target	Key activities in window
DOA+10	Project Plan + Deployment & Hosting Plan	DOA+7	Kickoff; baseline methodology agreed; OCIO environment coordination begins
DOA+21	UX wireframes, user flows, landing/nav redesign	DOA+18	Stakeholder sessions on live prototype; abandonment baseline captured
DOA+30	Technical Design Document (branching logic; low-code approach)	DOA+27	Schema + integration design confirmed with existing-system contractors

Clock (from DOA)	Milestone / Deliverable	Our internal target	Key activities in window
DOA+45	Alpha demonstration build	DOA+40	Branching form + landing on CDC infrastructure; ATD documentation started
DOA+90	Beta release + survey instruments, Phase 1 upload + Phase 2 provisions, draft ATD package, compliance package, ACR/VPAT	DOA+80	Full functionality; Government scenario testing; remediation window
Pre-deployment	ATD issued (PRS#12)	Beta+30	Review-comment remediation; pre-deployment vulnerability scan
DOA+150	Production-ready Final Release + source/build artifacts, user guides, O&M docs	DOA+140	Acceptance testing; defect burn-down; updated ACR
Monthly / bi-weekly	Status reports (10th of month); technical reviews; transition-readiness assessments	—	Continuous

Internal targets run 3–10 days ahead of every contractual clock; the margin is our schedule risk reserve. Bi-weekly Government technical reviews fall on the same cadence as code delivery to the CDC repository, so review is always of current, running software.

5. Challenges and Solutions

#	Challenge	Solution
1	ATD timing risk — Government review cycles are outside contractor control, and production deployment waits on ATD	Draft package with Beta (60+ days before needed); dedicated remediation window; PM-owned comment turnaround ≤5 business days; SA&A artifacts begun at kickoff, not Beta
2	Coordination with existing VAERS contracts — integration constraints surface late in multi-contractor environments	Named integration contact requested at kickoff; transmission format/channel confirmed in writing during 1.1; VAERS-compatible output validated against Government-furnished samples before Alpha
3	Abandonment baseline (≥30% reduction) depends on a measurable starting point	Baseline methodology co-signed with CDC at kickoff per Task 1.10; PRS#4 is measure-and-report with approved methodology; funnel analytics designed in from wireframe stage
4	Assistance-feature trust and safety — a public-facing AI feature in a vaccine-safety context invites misuse and misinformation risk	Deterministic safety gate ahead of any generative response (emergency, medical-advice, causality, and injection inputs receive fixed approved responses); CDC-approved content corpus; kill-switch configuration; full plan in Attachment 1 — and the guardrails are live and testable in the prototype today
5	PRA implications for surveys/form changes could stretch timelines outside the PoP	Flagged in our pre-quote questions; survey instruments delivered to clock regardless; schedule assumes CDC owns any required clearance in parallel [FLAG: confirm assumption wording after Bish Q&A response]

#	Challenge	Solution
6	Nine-month PoP with fixed clocks leaves no absorption room for a slow start	We start from working, tested software (Tab 2-2), a confirmed team (Tab 3-1), and internal targets ahead of every clock; week 1 is stakeholder sessions on a running system

Tab 2-2: Prototype

Prototype URL: <https://vaers-demo.frasierdigital.com> (no login required; desktop and mobile)

Frasier Digital submits a **working web application**, not a mockup. The prototype was built with the same open-source stack proposed in Tab 2-1 (React front end, schema-driven form engine) and demonstrates the core PWS capabilities on synthetic data. It is hosted outside Government infrastructure for evaluation convenience only; the proposed solution is developed and deployed entirely within CDC's FedRAMP-authorized Azure environment. The prototype collects, transmits, and stores no PHI/PII.

An in-application version of this walkthrough, with one-click sample scenarios (a completed parent report and the PRS#1 pharmacist error case), is at the **"For RFQ evaluators"** link in the site footer.

Suggested evaluation walkthrough (10 minutes)

1. **Public path:** Start a report as a member of the public. Try "Prefer to start by telling us what happened?": describe the event in plain words and the assistant proposes form answers drawn only from what you wrote — each individually reviewed, confirmed, and tagged "AI-suggested, verify" until you edit or accept it, with a receipt showing where answers landed. Note plain-language questions, "Why we ask" tooltips, inline validation, and the live completeness meter. Submit and inspect the structured record: field names mirror the published VAERS data dictionary (VAERSDATA / VAERSVAX), grouped to match the public-release de-identification split.
2. **Provider path (PRS#1):** Start over as a healthcare professional and select "*vaccine administration error with no adverse event.*" Every adverse-event question is suppressed and the form states how many questions were removed.
3. **Mobile:** Repeat either path on a phone.
4. **Low-code administration (PRS#8):** Open *Admin*, edit a field label, tooltip, or FAQ entry, and see the change live in the form — no developer involved.
5. **Uploads (PRS#9/#11):** Attach a PDF (accepted); attempt an image (rejected with Phase 2 explanation). At review, note auto-suggested supporting documents (PRS#10).

Feature-to-requirement map

Prototype feature (all working)	PWS	PRS
Branching submission form, public vs. provider; provider error-no-AE path suppresses all AE fields	1.6, 1.6.2	PRS#1
Plain-language public version with clinical/plain toggle	1.6.1	PRS#1
Full VAERS 2.0 data-element coverage (demographics through prior-event history); submission generates a record whose field names mirror the published VAERS data dictionary (VAERSDATA flags, VAERSVAX rows, computed NUMDAYS) with the public-release de-identification split	1.6, 1.9	PRS#5, #6
Mobile-first responsive; full flows complete on phone viewports	1.3	PRS#2

Prototype feature (all working)	PWS	PRS
Redesigned landing page: clear paths to reporting form, FAQs, data downloads	1.4	PRS#3
Intelligent completion assistance: smart inline validation, "Why we ask" tooltips, reactive FAQs, and a live AI assistant (form questions + narrative review, scripted fallback). Production-grade safety architecture: a classification stage routes safety-critical inputs (emergencies, medical-advice and causality questions, personal information, prompt injection) to fixed approved responses before any generative model runs, under a strict cannot-say policy. Production targets CDC's EDAV Azure OpenAI — no new AI services, Government environment only	Scope §2–3	—
Live completeness meter surfacing required/critical VAERS data elements to the submitter — the same metric CDC measures, driving the ≥90% completeness and abandonment-reduction targets	1.1, 1.10	PRS#4, #5
Medical-record upload + free-text box; Phase 1 type restrictions enforced; Phase 2 (images) accommodated by design	2.1, 2.2, 2.4	PRS#9, #11
Supplemental-document suggestion tool scanning provider submissions	2.3	PRS#10
Site-navigation survey + post-submission popup survey	1.5, 1.7	PRS#7
Low-code administration with live preview: content (labels, tooltips, FAQs, site text) AND interface (field visibility, choice options) — the PWS's "content and interface updates" demonstrated literally	1.8	PRS#8
Section 508: keyboard navigable, screen-reader labeled, WCAG 2.1 AA — automated audit passes with zero violations across all pages and form states	3.5	PRS#14
Performance: <1s loads (75 KB compressed application), save-and-resume	1.10	PRS#4

Architecture and scalability

The form is not hard-coded: sections, fields, branching conditions, plain/clinical text variants, tooltips, and validation rules are a **declarative configuration schema** rendered by a generic engine. One design decision serves four evaluation areas: branching correctness is testable data, not scattered code (PRS#1); the low-code interface edits the same configuration the form renders (PRS#8); VAERS field mappings live beside the fields they map (PRS#6); and future changes — new vaccines, Phase 2 image upload, new submitter types or languages — are configuration, not redevelopment. The demo persists configuration locally; production uses a role-restricted, versioned, audit-logged configuration service in CDC Azure. **Risks demonstrated and mitigated:** AI-scope ambiguity is hedged (validation, tooltips, and FAQs work with no AI service; the assistant is an additive layer); Phase 1/2 upload separation is enforced exactly as the PWS directs; plain-language substitution is a content operation CDC staff perform themselves, mitigating PRA-driven wording turnaround.

Tab 3-1 — Management Plan

1. Organizational Structure and Program Manager Authority

Frasier Digital is a principal-led small business: the Program Manager is the company's founder and managing member, and he personally leads this delivery. That structure is CDC's management-risk mitigation. There is no account layer, no matrixed resource pool, and no approval chain above the PM — **the person accountable to the COR has full, immediate authority over staffing, budget, priorities, and corrective action**, and the distance between a Government comment and a binding decision is one phone call.

Joseph Frasier (PM / Lead Architect, Key Personnel) directs the program, owns all deliverables, chairs the bi-weekly technical reviews, and is the single point of contact for the CO and COR. **Ryan Daley (Technical Lead, Key Personnel)** owns day-to-day engineering execution and release quality; his healthcare-IT background (Optum/ UnitedHealthcare; Shuttle Health) anchors the team's PHI/PII and ADA-conformance discipline. Supporting engineers report to Ryan; content and UX-research support report to Joseph. The team communicates in a single shared channel with CDC-visible status — no internal information asymmetry for the Government to penetrate.

Escalation path: any team member → PM within the same business day; PM → COR per the communication plan below. For incidents, the 1-hour reporting channel operates independently of the management chain so notification is never queued behind analysis.

2. Staffing Plan

The team is named, confirmed, and sized honestly for a nine-month FFP delivery (~1.8 FTE average, weighted toward the build phase; commitments below reconcile exactly with the labor hours itemized in Volume II):

Person	Role	Commitment	Relevant depth
Joseph Frasier	PM / Lead Architect + Security & Compliance Lead (KEY)	0.55 FTE (both roles)	Federal delivery lead; architecture; prototype author; SA&A/ATD/EPLC artifact ownership
Ryan Daley	Technical Lead — healthcare IT (KEY)	0.25 FTE	Optum/UnitedHealth; Shuttle Health; ADA-conformant UI delivery; CI/CD ownership
Rahmin Shoukoohi	Integration & data engineer	0.15 FTE	HolonHealth (60k-user healthcare platform, auditable ledger); state education data-exchange; Atlanta-area — aligned to CDC core hours and available for coordinated on-site presence
Andrew Frasier	Frontend engineer, test automation, 508 testing	0.50 FTE	Component build-out; automated accessibility testing in CI
Seth Chesky	QA & accessibility engineer	0.20 FTE	Release-candidate QA cycles; manual assistive-technology testing; current on our component stack

Person	Role	Commitment	Relevant depth
Rida Khazi	Plain-language content, UX research, satisfaction surveys	0.12 FTE	Public-path language (Task 1.6.1); survey instruments (1.5/1.7)

Rapid staffing. Our staffing risk is managed by design rather than by requisition: every person above is identified by name today, has worked with this team's stack and standards, and requires no recruiting pipeline. A pre-identified bench of additional vetted engineers from our standing roster can be activated inside two weeks — onboarding for this contract means Tier 2 (MRPT) paperwork, NDAs, and CDC trainings, not finding a stranger. Roster submission 30 days before the effective date, background investigations, and training compliance are managed by the PM as contract deliverables with the same tracking as technical milestones.

Capacity disclosure. Frasier Digital currently has proposals under evaluation with other agencies with periods of performance that could overlap this one. We size and price every bid against total committed capacity: the FTE commitments above are net of all potential concurrent awards, the named bench exists to absorb surge, and the PM's combined 0.55 FTE on this contract is protected as a floor, not an average. We flag this proactively because deliverable-based FFP work only functions when capacity claims are honest. **[FLAG: keep wording synced with actual award status at render time — mirrors NOAA disclosure pattern.]**

Employee turnover — past three years. Frasier Digital does not operate a salaried-staff pool, so we answer this requirement transparently rather than statistically: delivery is performed by the firm's principal together with named associates who are individually committed to a specific engagement at proposal time and engaged under written agreement at award. No Frasier Digital delivery-team member has ever departed during an engagement.

Year	Salaried technical employees	Departures (voluntary/involuntary)	Mid-engagement departures from delivery teams
2024	0	0 / 0	0
2025	0	0 / 0	0
2026 (YTD)	0	0 / 0	0

On a nine-month contract, the turnover risk that matters is not a statistic but a question: *will the people named in this proposal be the people doing the work?* Our answer is structural, not historical — key personnel are contractually committed to this engagement, substitution requires CO/COR approval under the contract terms, each named associate has confirmed availability for this period of performance before being proposed, and the accountable principal cannot leave his own firm. A named, pre-identified bench (Section 2) backstops the commitment rather than a recruiting pipeline.

3. Performance Metrics and Measurement

Our internal metrics mirror the Government's Performance Requirements Summary line for line, measured continuously in CI and reported monthly — so Government acceptance testing confirms numbers we already track rather than discovering new ones:

PRS focus	Our internal metric and cadence
PRS#1 branching correctness	Scenario-matrix pass rate (100% target) — every commit, in CI
PRS#2 responsiveness	Device/browser matrix regression — every release candidate
PRS#3 landing/navigation	Access-path functional checks — every release
PRS#4 performance	Real-user page-load p95, submission-time median, abandonment funnel — continuous, reported monthly with methodology
PRS#5 data completeness	Required/critical field completeness on synthetic then live reports ($\geq 90\%$ target) — weekly analysis
PRS#6 VAERS-compatible output	Structured-output validation against data-dictionary mappings — every commit
PRS#7 surveys	Deployment status + response-data delivery per approved schedule
PRS#8 low-code editability	Government update-scenario rehearsal pass rate (≤ 1 dev intervention) — monthly from Alpha
PRS#9–11 upload/Phase 1	Upload defect counts by severity; Phase 1 restriction enforcement tests — every release
PRS#12–13, #15 security/ATD	SA&A artifact status; scan findings burn-down (zero known critical/high at deployment) — bi-weekly
PRS#14 Section 508	Automated axe pass (0 violations) every commit + manual AT testing every release; ACR currency
PRS#16 documentation	Guide completeness vs. final UX — at Beta and Final
PRS#17 PM reporting	On-time PM deliverable rate ($\geq 95\%$); bi-weekly review attendance (100%)
PRS#18 transition readiness	Monthly readiness assessment delivered; repo/docs currency check

Metrics appear in every monthly status report with trend lines, threshold status, and — where a threshold is at risk — a named corrective action with an owner and a date.

4. Quality Assurance, Risk Management, and Communications

Quality control. Quality is enforced where it is cheapest — in the pipeline, before Government review: author-*≠*-reviewer code review on every change; CI gates (unit, integration, accessibility, cross-viewport, structured-output validation) that block merge on failure; bi-weekly code delivery to the CDC repository so the Government always reviews current, running software; severity-classified defect triage with remediation SLAs through Beta and Final. The QA function is owned by the Technical Lead and audited by the PM — findings, not assertions, drive release decisions.

Risk management. A living risk register (established at kickoff, reviewed at every bi-weekly technical review, summarized monthly) tracks each risk with probability, impact, trigger, owner, and mitigation. The top program risks and their mitigations are detailed in Tab 2-1 §5 — notably ATD review timing (draft package 60+ days early with a dedicated remediation window), multi-contractor integration (written interface confirmation during Task 1.1), and the abandonment

baseline (methodology co-signed at kickoff). New risks enter the register through anyone on the team; only the PM retires them.

Change control. Scope, architecture, and performance standards are protected by a formal change process: any proposed change is written up with cost/schedule/performance impact, reviewed by the PM, and — where it touches contract scope or Government-approved baselines — submitted to the CO/COR for approval before implementation. Configuration changes made through the Task 1.8 low-code interface are versioned with one-click rollback, so CDC's own content changes are governed by the same discipline without developer gatekeeping.

Communications. Bi-weekly technical reviews chaired by the PM against working software; monthly status reports (by the 10th) covering accomplishments, schedule, risks, staffing changes, and planned-vs-actual expenditure; a Cost Management Plan maintained and current per Subsection C; ad-hoc access to the PM during CDC core hours (8:30–5:00 ET) with same-business-day response; 1-hour incident reporting. We operate on a no-surprises rule: any threshold at risk is in front of the COR before it is a problem, with a proposed correction attached.

5. Subcontractor Management

Frasier Digital will perform this requirement entirely with its own personnel. **No subcontracting opportunities exist under this effort**; accordingly, no subcontractor management plan or draft subcontract agreement is required. Should the Government later direct or approve a change to this posture, subcontracted work would be managed under the same QA gates, metrics, and communication cadence described above, with a draft agreement provided for Government review before execution.

Tab 4 — Similar Experience

1. Anchor Engagement — Region 4 Education Service Center (Houston, TX)

Certification training and completion-tracking platform, built from scratch and operated at scale. July 2024 – October 2025 · total engagement approximately \$2 million · Frasier Digital principal-delivered (PM + development). [FLAG: \$2M figure conditional on reference verification per consistency map — fallback wording "a multi-year, seven-figure engagement" if unconfirmed at render.]

Region 4 ESC engaged Frasier Digital to replace manual certification processes with a statewide-scale online platform serving school districts across all of Harris County and every additional entity dependent on Region 4's SCORM-based training system. Frasier built the entire system with no inherited codebase: full authentication backend, SCORM Cloud API integration, course loading/ingestion and validation, and end-to-end completion tracking — delivered and operated for **thousands of concurrent non-technical users** (school bus drivers, teachers, district staff).

The parallel to this requirement is structural, not cosmetic:

Region 4 (delivered)	VAERS modernization (required)
Public-facing workflows for non-technical users at population scale	Public + provider reporting for the general population
Guided flows, validation, and error recovery to protect data quality	Branching form, validation, intelligent assistance (Tasks 1.3–1.6)
Completion tracking with structured records feeding institutional reporting	VAERS-compatible structured capture and transmission (Task 1.9)
Privacy-sensitive education records context	PHI/PII HIPAA/Privacy Act context (Task 3)
Integration with an incumbent ecosystem (SCORM Cloud, district systems)	Integration with existing VAERS systems and contracts
Built from scratch to production under a small named team	9-month design-build-deploy with fixed clocks

At roughly four times the size of this requirement's anticipated value, Region 4 demonstrates the firm delivering the same shape of work — new-build, public-facing, data-quality-critical, integration-constrained — at greater scale than CDC is buying here.

Reference (customer POC): Ethan Gula, Software Developer II, Region 4 ESC · (832) 585-3947 · ethan.gula@esc4.net **[FLAG: swap/augment with durable contact + backup Region 4 POC before render — evaluation runs past the POC's Region 4 tenure.]**

2. Healthcare-IT Domain Expertise (Named Delivery Team)

The RFQ emphasizes healthcare-IT domain expertise. Ours is carried by the named people who will perform, not a corporate claims sheet:

Team member (role on this contract)	Healthcare-IT experience
Ryan Daley (Technical Lead, KEY)	Optum / UnitedHealthcare: engineered ADA-compliant member-facing UIs on Azure with enterprise CI/CD. Shuttle Health (healthtech): micro-frontend platform architecture, AWS serverless services, 90%+ automated test coverage — production healthcare software discipline
Rahmin Shoukoohi (Integration & data engineer)	HolonHealth (current): senior engineer on a healthcare rewards platform serving 60,000+ clients; designed its ground-up auditable points-ledger — audit-grade data integrity in a healthcare compliance context. State education agency (2024–present): led 3-person team, 20+ secure data-exchange features, 69 districts
Joseph Frasier (PM / Lead Architect, KEY)	Principal experience architecting and delivering enterprise platforms including current Fortune 100–500 AI implementations (full-stack, end-to-end) and production generative-AI features; regulated-data discipline across education (FERPA-adjacent) and government delivery

Collectively: member-facing healthcare UI at national-payer scale, healthcare platform data integrity, and PHI/PII-adjacent engineering habits — resident in the exact people staffed on this contract (Tab 3-1).

3. Documented Evidence

- **Working prototype of this requirement** — <https://vaers-demo.frasierdigital.com> (Tab 2-2): the most direct evidence offered — similar experience made inspectable, built on the VAERS 2.0 data elements with 0 WCAG 2.1 AA violations across all tested states.
- **Region 4 project summary and reference POC** (§1 above); the reference is available for Government feedback per the RFQ's past-performance provisions.
- **Federal delivery artifacts on request:** recent federal proposal deliverables and past-performance questionnaire responses completed by the Region 4 customer. **[FLAG: confirm we want this line — it invites the ask; delete if not.]**
- **Engineering evidence embedded in the prototype:** automated accessibility audit results, end-to-end test suite, and open-source-ready codebase demonstrating the CI discipline described in Tabs 2-1/3-1.

Tab 3-2 — Key Personnel Resumes

Joseph Frasier — Program Manager / Lead Architect (Key Personnel)

Frasier Digital, LLC · Tomball, TX · joseph@frasierdigital.com · (850) 356-2382

Summary

Founder and principal of Frasier Digital with 11+ years of software engineering and delivery leadership across government-facing, education, and enterprise platforms. Program manager and lead architect on the proposed VAERS modernization: personally architected and built the working prototype submitted with this quote. Career pattern matches this contract's shape — new-build, public-facing, data-quality-critical applications delivered by small named teams under fixed schedules, with privacy-sensitive data handled correctly from the first commit.

Role on this contract

Program Manager / Lead Architect and Security & Compliance Lead (combined 0.55 FTE, key personnel): single point of contact for the CO/COR; owns architecture, deliverable schedule, the SA&A/ATD documentation path and EPLC stage gates, risk register, and bi-weekly technical reviews. Author of the Tab 2-2 prototype.

Relevant experience

Frasier Digital, LLC — Founder & Managing Member / Program Manager · 2024–present

- **Region 4 Education Service Center (Houston)**, Jul 2024–Oct 2025: PM and developer on a certification training platform built from scratch — SCORM Cloud API integration, full authentication backend, course ingestion and validation, completion tracking — serving thousands of non-technical users across all Harris County school districts and other entities dependent on Region 4's training system. No inherited codebase; delivered to production and operated. **[FLAG: engagement value reference per Tab 4 rule]**
- Federal proposal and prototype delivery practice: working evaluation prototypes with live AI-assist features behind deterministic safety guardrails, automated 508 testing in CI, open-source posture per M-16-21.

PwC — Senior Software Engineer (AI implementations) · 2026–present

- End-to-end AI implementations for Fortune 100–500 clients: full-stack build, solution architecture, design, CI/CD, and maintenance. Production discipline for LLM-backed features: structured outputs, guardrails, evaluation, cost controls — directly applied to this contract's intelligent-assistance layer.

Louisiana Department of Education (via Insight Global) — Senior Software Engineer · Jul 2024–present

- Maintains and extends C++ computational models at the core of Louisiana's statewide education data processing (performance, memory, numerical correctness), plus the surrounding .NET/C#/SQL Server platform and T-SQL data pipelines — statewide-scale government data quality as a daily job.

Perdoceo Education Corporation (via DigitalCrafts) — Lead Instructor / Senior Software Developer · Jan 2021–Jan 2024

- Canvas LMS engineering across the Perdoceo enterprise: integrations, automation, and platform work for a national online-education operation.

Mailchimp (via Tensure) — Software Engineer, Generative AI · Jan–Jun 2024

- Shipped production generative-AI features at consumer scale.

Deloitte — Software Engineer · Apr 2020–Jan 2021

- USDA-facing crop-inspection scheduling application (mobile + desktop) — federal field-user UX and delivery cadence.

Earlier engineering foundation · 2015–2020

- **EOG Resources** (Sep 2018–Apr 2020): company-wide water/gas/oil production monitoring and forecasting platform; front end plus C++ water-production model integration; Mapbox scenario visualization; onboarded 50+ users.
- **WeWork** (Apr 2019–Apr 2020): React/TypeScript internal and customer-facing apps.
- **Ecotone** (Jan 2015–Sep 2018): front-end JavaScript/React, Node/Express.
- Concurrent engagements presented per-engagement.

Skills

TypeScript/JavaScript, React, Node.js, C++, C#/.NET, SQL (SQL Server/PostgreSQL), T-SQL pipelines; Azure + AWS; CI/CD (GitHub Actions); LLM integration with structured outputs and safety gating; Section 508/WCAG engineering; SCORM/LMS ecosystems; federal deliverable and compliance documentation (EPLC, SA&A artifact preparation).

Education

B.S. Business Management (minor, Business Technology), Pensacola Christian College, 2013 — including three years of computer science major coursework. The Iron Yard immersive software engineering program, 2017.

Ryan Daley — Technical Lead, Healthcare IT (Key Personnel)

Houston, TX · ryandaley321@gmail.com · (832) 998-0924

Summary

Technical lead and senior software engineer with 7+ years building production web platforms, including member-facing healthcare applications at national-payer scale (Optum/UnitedHealthcare) and healthtech platform engineering (Shuttle Health). Owns architecture, CI/CD, and release quality on rapid-delivery teams; testing discipline to 90%+ automated coverage; accessibility (ADA/508) built into the pipeline rather than audited afterward. On this contract: day-to-day engineering lead accountable for release quality across Alpha, Beta, and Final.

Role on this contract

Technical Lead — healthcare IT (0.25 FTE, key personnel): owns engineering execution, code review standards, CI/CD gates (unit, integration, accessibility, cross-viewport), defect triage SLAs, and bi-weekly build delivery to the CDC repository.

Relevant experience

PwC (Vanguard team) — Senior Software Engineer · Apr 2025–present

- Rapid AI-integrated prototypes for Fortune 500 clients: React/Next.js/TypeScript with LLM integration; owns CI/CD for the team. The prototype-to-production cadence this contract's 21/45/90/150-day clocks demand, practiced weekly.

Shuttle Health — Senior Software Engineer (healthtech) · 2022–2024

- Micro-frontend platform architecture for a healthcare startup; AWS API Gateway, Lambda, S3, Elasticsearch; drove automated test coverage above 90%. Healthcare product engineering end-to-end: data sensitivity, reliability, and iteration speed together.

Optum / UnitedHealthcare — Software Engineer · 2021–2022

- Built **ADA-compliant member-facing UIs** at national-payer scale on Azure with Jenkins/Terraform pipelines — direct precedent for this contract's 508-conformant, Azure-hosted public experience.

Bank of America — Software Engineer · 2020–2021

- React front ends with Java Spring Boot services in a regulated, review-gated enterprise environment.

IBM — Software Engineer · 2019–2020

- Real-time operations dashboards; Watson NLP integration.

Stellar Digital — Co-founder · 2017–2018

- Client web delivery; end-to-end ownership from requirements to production.

Skills

TypeScript/JavaScript, React, Next.js, Node.js, GraphQL; PostgreSQL, MySQL, MongoDB, Elasticsearch; Azure and AWS; Kubernetes, Docker, Terraform; GitHub Actions and Jenkins CI/CD; testing with Vitest/Jest/Playwright; accessibility engineering (ADA/WCAG/508); secure SDLC in regulated environments.

Education

Flatiron School, software engineering immersive, 2019. The Iron Yard, software engineering immersive, 2017.